Mikael GALLEGO

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EDUCATION

Université de Bordeaux | *Doctorat Biophysique* 2026 — *Present*
Université de Bordeaux | *Master Physique fondamentale et applications* 2024 — 2026
NC State University (NCSU) | *B.S. Physics, B.S. Mathematics* 2020 — 2024

COURSEWORK

- Numerical Analysis, Physics Simulations
- Quantum Mechanics, Quantum Optics
- Electrodynamics
- Biophysics, Soft Matter Physics
- Nanophysics, Nanophotonics
- Laser Physics and Nonlinear Optics

SKILLS

- **Computer languages and software:** Python, Wolfram, C, LaTeX, Microsoft Office, Linux/UNIX, OSLO, MPI, IBM Spectrum LSF, LabView.
- **Human languages:** English (native), French (C1).

EXPERIENCE

Laboratoire Ondes et Matière d'Aquitaine (LOMA) | *Intern* Feb. 2026 — Jul. 2026
- Characterized 'gliding' motility of the *C. reinhardtii* microalga.
- Applied time-frequency analysis methods like the wavelet transform to real human EEG data to analyze sleep activity in the brain. Wrote Python algorithms to filter and detect sleep 'spindles' in large datasets.

Integration from Material to System Laboratory (IMS) | *Intern* Jan. 2025 — July 2025
- Designed and built a pump-probe setup for an array of terahertz-emitting nanoantennae to be used for non-destructive sensing in biomedical applications.

Nonlinear Artificial Intelligence Lab (NAIL) | *Research Assistant* Feb. 2023 — Feb. 2026
- Investigated the effect of spatial disorder 'taming' chaotic systems of nonlinear pendula under full ranges of parameter values. Results published in Physics Letters A.

NCSU Physics Senior Laboratory | *Student* Aug. 2023 — Dec. 2023
- Designed and built an optical system for the creation and detection of correlated photon pairs.

NCSU Physics Department | *Employee* April 2022 — July 2024
- Constructed classroom demonstrations for physics and engineering professors.

PUBLICATION

- **M. Gallego**, J. F. Lindner, A. Radhakrishnan, W. L. Ditto, Lyapunov landscape elucidates disorder-taming-chaos in arrays of coupled pendula, Physics Letters A 579 (2026) 131483. DOI: 10.1016/j.physleta.2026.131483

CONFERENCES

- Presented: *"Photoconductive terahertz antenna source array for nondestructive testing"*, EUR LIGHT Master Workshop, Université de Bordeaux, June 2025.
- Presented: *"Disorder Taming Chaos in Coupled Pendulum Arrays"*, Undergraduate Research Symposium, NC State University, April 2024.
- Presented: *"Single Photon Quantum Mechanics"*, Senior Physics Lab Presentations, NC State University, December 2023.